Clustering, etc. in $G_{n,p}$

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Triangles in $G_{n,p}$

To make a triangle, we need 3 nodes, and we need all 3 possible edges between them to exist. For a graph with n nodes there are $\binom{n}{3}$ distinct sets of 3 points. The probability that all three edges of the triangle exist is p^3. If $T(G)$ is the number of triangles in G then on average we should expect

$$\# \textit{ of triangles in } G_{n,p} \approx \binom{n}{3} p^3 = E[T(G_{n,p})]$$

The probability that k triangles exist is $Binomial(T(K_n), p^3)$, that is

$$Pr\{T(G_{n,p}) = k\} = \binom{T(K_n)}{k} p^{3k}(1-p^3)^{T(K_n)-k}$$

where K_n is the complete graph on n nodes. Noting that $T(K_n) = \binom{n}{3}$ the probability mass function for the number of triangles, $k \in \{0, 1, ..., \binom{n}{3}\}$, in the random graph $G_{n,p}$, is

$$f(k) = \binom{\binom{n}{3}}{k} p^{3k}(1-p^3)^{\binom{n}{3}-k}$$

2-Paths ('biangles') in $G_{n,p}$

Let the term '2-path' denote a set of 3 vertices connected by 2 edges. For brevity we'll use the word biangle to denote these objects. Since for each set of 3 points there are 3 possible biangles, the number of possible biangles is $3\binom{n}{3}$. The probability that both edges in a particular biangle exist is p^2. If $\angle(G)$ is the number of biangles in G then on average we should expect

$$\# \textit{ of biangles in } G_{n,p} \approx 3\binom{n}{3} p^2 = E[\angle(G_{n,p})]$$

Following a similar line of reasoning as in the triangle case, we can write down the probability mass function for the number of biangles, $m \in \{0, 1, ..., 3\binom{n}{3}\}$, in $G_{n,p}$ as

$$Pr\{\angle(G_{n,p}) = m\} = \binom{\angle(K_n)}{m} p^{2m}(1-p^2)^{\angle(K_n)-m}$$

or

$$g(m) = \binom{3\binom{n}{3}}{m} p^{2m}(1-p^2)^{3\binom{n}{3}-m}$$

Clustering Coefficient of $G_{n,p}$

We will define the clustering coefficient of a graph G as 3 times the number of triangles divided by the number of biangles,

$$C(G) = 3\frac{T(G)}{\angle(G)}$$

The factor of 3 is included because for each triangle we have 3 biangles, and we would like 'clustering' to be a measure of the percentage of possible triangles that are present in our graph. To illustrate, consider the complete graph K_n, where

$$C(K_n) = 3\frac{T(K_n)}{\angle(K_n)} = 3\frac{\binom{n}{3}}{3\binom{n}{3}} = 1$$

which is consistent with the idea that the complete graph is maximally clustered.

It doesn't seem like a quick task to write down the probability mass function for C. However, it is clear by a heuristic argument that $E[C] = p$. The probabilities of edges a and b existing are independent, so for every biangle present the probability that the third edge of the triangle exists is just p, and this should be the fraction of all possible triangles that are present, which is exactly what we were trying to define by 'clustering coefficient'. While the result that $E[C] = p$ does seem to be correct, this argument fails to give us a means for calculating $V[C]$, etc.

Since $C = 3T/\angle$ one might be tempted to say

$$E[C] = 3E\left[\frac{T}{\angle}\right] = 3\frac{E[T]}{E[\angle]}$$

but the last equality is not true in general since T and $\angle$ are not independent random variables . In particular, we know that $\angle \geq 3T$ (i.e. the support is not a product space).

Numerical Results

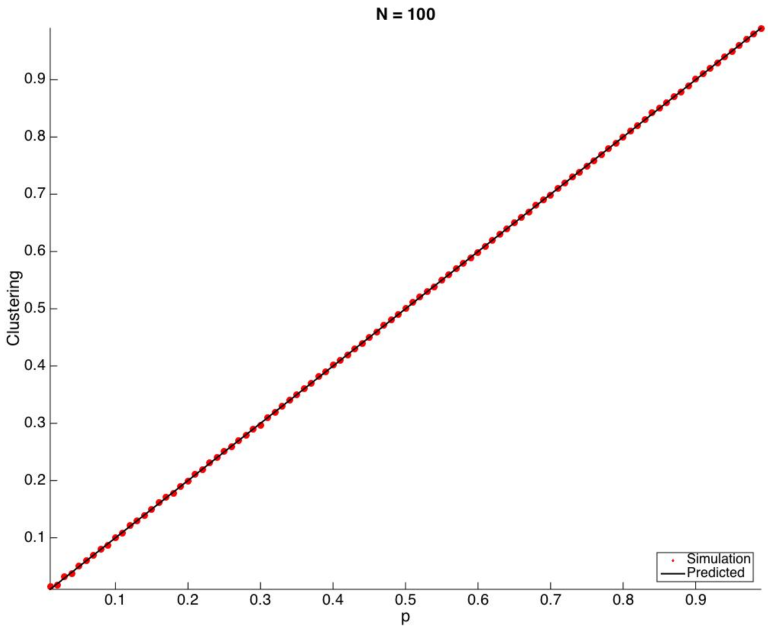
To calculate the clustering coefficient for a realized graph G we have

$$C(G) = 3\frac{\text{\# of triangles}}{\text{\# of biangles}}$$

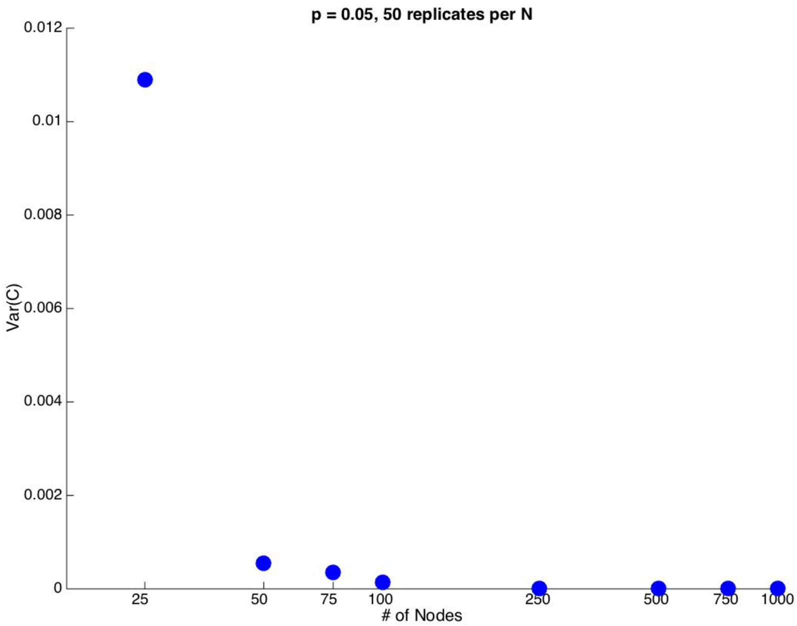
which can be computed using the adjacency matrix A of G,

$$C(G) = \frac{\sum_i \lambda_i^3}{e^T A^2 e - Tr(A^2)}$$

where λ_i are the eigenvalues of A and e is a column vector of ones. To approximate the expected value of C we generate an ensemble of 50 graphs for each value of $p \in \{0, 0.01, 0.02, ..., 0.99, 1\}$. We then look at the mean and variance of the clustering coefficient in each ensemble. We find that $E[C] = p$ as expected, with a variance that seems to vanish as $N \to \infty$. Results shown below are for $N = 100$.



To support the idea that the variance vanishes (and hence the random variable C converges to a degenerate distribution for large N), we fix a p value, look at a sequence of increasing N values, create an ensemble of 50 graphs for each, and plot the variance. Results for $p = 0.05$ are shown below for graphs with 10 to 1000 nodes.


p = 0.05, 50 replicates per N
0.012
0.01
0.008
0.006
0.004
0.002
0
Var(C)
25
50
75
100
250
500
750
1000
# of Nodes